IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2019

EEE PART I: MEng, BEng and ACGI

Corrected Copy

a) Question p1
Q1 part a) with 4 marks
(announced before exam + written on white board)

b) Fig 3.1 Vout :
connected to Q5 drain
Q6 drain

ANALOGUE ELECTRONICS 1

Tuesday, 4 June 10:00 am

Time allowed: 2:00 hours

There are THREE questions on this paper.

Answer ALL questions.
Q1 carries 40% of the marks. Questions 2 and 3 carry equal marks (30% each).

Any special instructions for invigilators and information for candidates are on page 1.

Examiners responsible First Marker(s) : P.D. Mitcheson
Second Marker(s) : C. Papavassiliou

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The Questions

1. For each part of this question, clearly state any assumptions you make.

a) Explain why small-signal modelling and analysis is a useful tool in analogue electronics.

[3]

b) What is the operating mode of the MOSFET in Figure 1.1? What can you say about the length of the inverted channel region compared to the distance between the drain and source diffusions? Explain your answers.

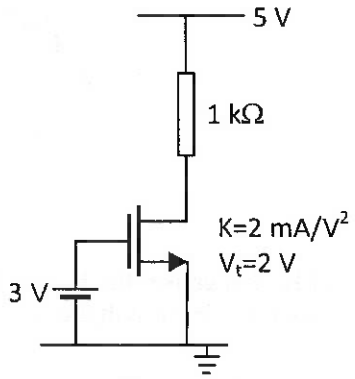


Figure 1.1

[6]

c) Show that for the common emitter amplifier in Figure 1.2 the small-signal gain is independent of the value of R_C assuming the quiescent output voltage is $V_{cc}/2$. You may neglect the transistor's output resistance. Suggest two ways of increasing the gain.

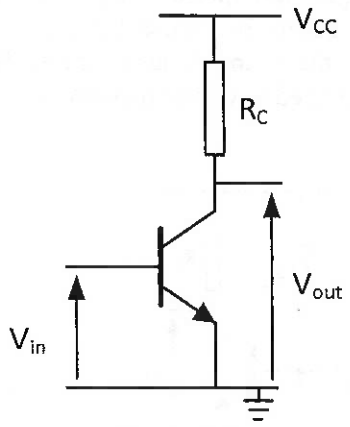


Figure 1.2

[6]

d) With the aid of a diagram, explain why the BJT-based class B output stage suffers from cross-over distortion.

(4)

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[4]

e) The circuits of Figures 1.3 and 1.4 are proposed for use as logic inverters. A logic low input is 0 V and a logic high input is 5 V. A logic high output is 5 V and a logic low output is a maximum of 0.2 V for both circuits.

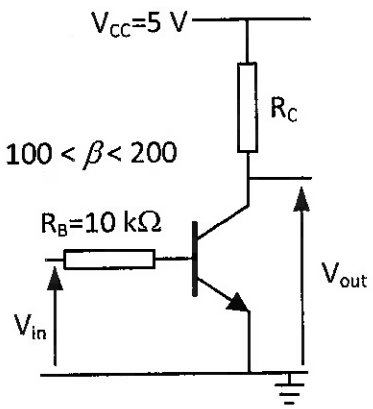


Figure 1.3

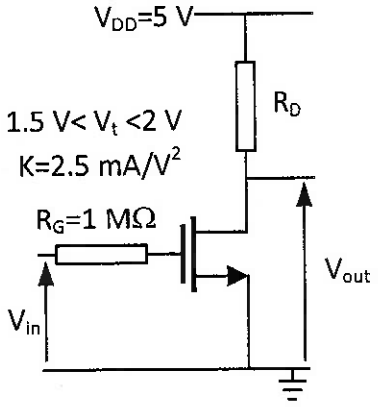


Figure 1.4

i) What is the value of R_c that causes the BJT to be overdriven by a factor of no more than 5 when the input voltage is 5 V?

[6]

ii) Does the value of R_c calculated above allow the inverter to work to specification across the entire range of transistor tolerances?

[2]

iii) What is the minimum value of R_D that meets the specifications for the inverter for all possible MOSFETs? Ignore channel length modulation.

[4]

f) Describe the advantages and disadvantages of each of the BJT amplifier configurations shown below in Figure 1.5(a-c). Include in your answer a comparison between their small-signal gain, bias stability, and input resistance. There is no need to derive mathematical formulae.

[9]

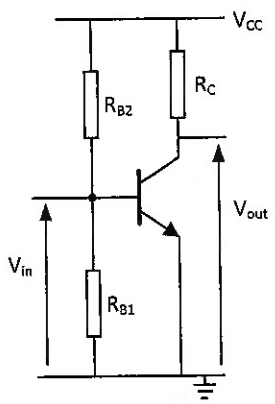


Figure 1.5a

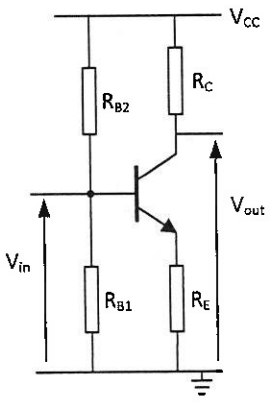


Figure 1.5b

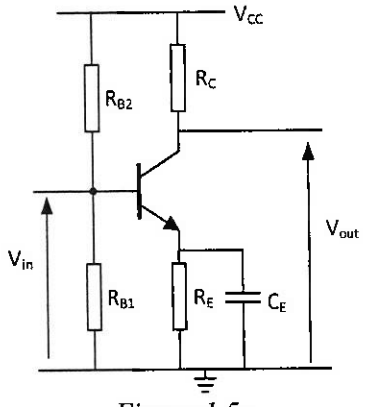


Figure 1.5c

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2. Figure 2.1 shows a single stage common emitter amplifier with an active load constructed from a simple current mirror. All transistors are matched and V_{CC} is 8 V.

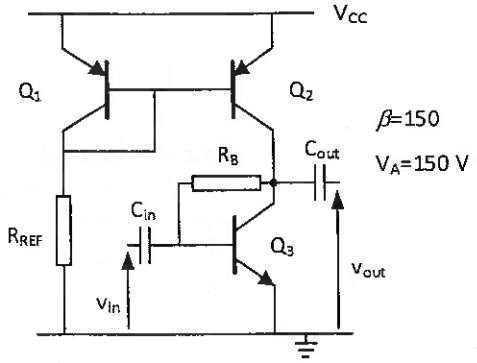


Figure 2.1

a) Neglecting finite beta error in the current mirror, choose a value of R_{REF} to bias the collector current of Q_1 at 1 mA and then choose a value of R_B that biases Q_3 to give a quiescent output voltage to give maximum head- and leg-room for the amplifier.

[6]

b) Draw a small-signal model of the amplifier, and assuming that at the frequency of operation C_{in} and C_{out} are short circuits, calculate the input and output impedance of the amplifier, and its small-signal gain. You may assume R_B is very large.

[8]

c) Draw a labelled small-signal macro-model of the amplifier.

[4]

d) It is proposed to cascade two of these amplifiers together with AC coupling to achieve a higher gain. Using the macro model from part c), what is the gain of the resulting circuit when driving a 10 kΩ load from a perfect voltage source? Assume all capacitors are short circuits at the operating frequency. Comment on your answer.

[6]

e) It is then proposed to cascade two amplifier stages together with DC coupling by omitting any capacitors between the two stages as shown in Figure 2.2. Explain what changes need to be made to the quiescent condition of the first stage amplifier and why the emitter follower stage (comprising Q_7 and R_E) has been added.

[6]

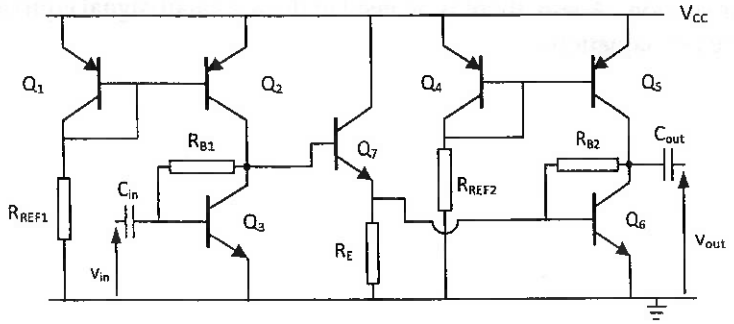


Figure 2.2

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3. Figure 3.1 shows a differential amplifier constructed using MOSFETs. Throughout this question assume all MOSFETs are matched.

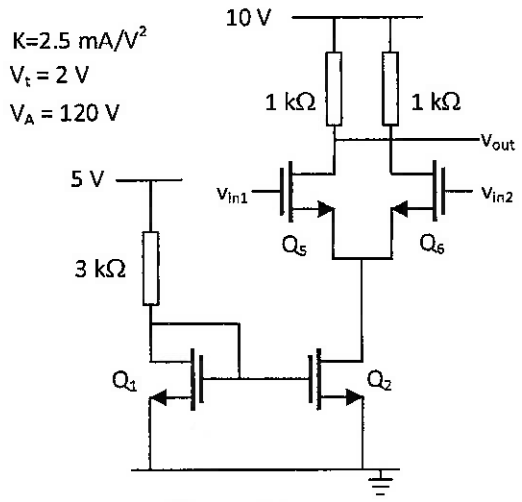


Figure 3.1

a)

i) Calculate the tail current set up by the current mirror comprising Q_1 and Q_2.

[3]

ii) Draw a small-signal model of the circuit and calculate the small-signal differential gain for a single-ended output (where $v_{in}=v_{in1}-v_{in2}$).

[9]

b) By assuming changes in v_{GS} of Q_5 and Q_6 are negligible, calculate the approximate values of the common mode gain and the common mode rejection ratio (CMRR).

[8]

c) It is proposed to improve the CMRR by replacing the simple MOSFET current mirror with a three-transistor Wilson current mirror of Figure 3.2a. Explain why the output impedance of this current mirror is superior to that of the simple current mirror and why this improves the CMRR. There is no need to draw a small-signal model nor derive any equations.

[6]

d) The four-transistor version of the Wilson current mirror is shown overleaf in Figure 3.2b. From your knowledge of the characteristic curves of a MOSFET, explain why this current mirror achieves current gain closer to 1 than the three-transistor version. Again, there is no need to draw a small-signal equivalent model nor derive any equations.

[4]

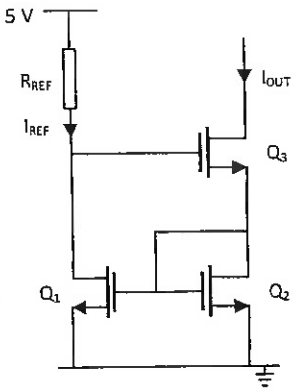


Figure 3.2a

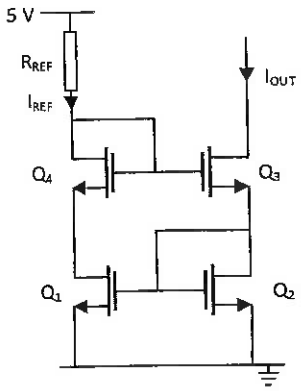


Figure 3.2b

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